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United States Patent	12415488
Kind Code	B2
Date of Patent	September 16, 2025
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Dynamic braking torque split for optimal thermal management

Abstract

Typically, mobile equipment with axles has a mechanically fixed or constant split ratio for the braking torque that is applied to each axle. Disclosed embodiments dynamically adjust the braking torque ratio between axles based on real-time parameter values, such as requested braking power and a real-time brake state parameter (e.g., brake temperatures), to more evenly distribute wear or other health imbalances across the brake systems of mobile equipment. Accordingly, disclosed embodiments may extend the longevity of brake systems, reduce the costs of maintenance of mobile equipment, facilitate a more cost-effective brake system that balances health or durability with performance under different operating scenarios, and/or the like.

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Appl. No.:	18/203490
Filed:	May 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240400015 A1	Dec. 05, 2024

Publication Classification

Int. Cl.: B60T8/26 (20060101); B60T8/171 (20060101); B60T8/172 (20060101); B60T8/1766 (20060101); B60T8/58 (20060101); B60T17/22 (20060101); F16D66/00 (20060101)

U.S. Cl.:

CPC **B60T8/26** (20130101); **B60T8/171** (20130101); **B60T8/172** (20130101); **B60T8/1766** (20130101); **B60T8/58** (20130101); **B60T17/22** (20130101); B60T2250/04 (20130101); F16D2066/001 (20130101); F16D2066/005 (20130101)

Field of Classification Search

CPC: B60T (8/171); B60T (8/172); B60T (8/1766); B60T (8/26); B60T (8/58); B60T (17/22); B60T (2250/04); F16D (2066/001); F16D (2066/005)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6719379	12/2003	Crombrez	N/A	N/A
7694555	12/2009	Howell et al.	N/A	N/A
10752288	12/2019	Chennupalli et al.	N/A	N/A
11590848	12/2022	Carbone et al.	N/A	N/A
11597283	12/2022	Suzuki et al.	N/A	N/A
2002/0030408	12/2001	Niwa	903/918	B60K 17/354
2003/0234577	12/2002	Crombez	303/156	B60W 10/18
2005/0137060	12/2004	Kuras	477/214	B60L 7/22
2008/0100129	12/2007	Lubbers	303/113.1	B60L 7/18
2008/0236269	12/2007	Howell	73/121	B60T 17/221
2009/0326770	12/2008	O'Neil	477/118	B60T 7/12
2015/0328992	12/2014	Amaravadi	701/70	B60L 7/22
2017/0057476	12/2016	Schwartz	N/A	B60T 8/172
2019/0211529	12/2018	Vigholm	N/A	B60T 8/36
2019/0299949	12/2018	Ying	N/A	B60T 8/1761
2019/0322316	12/2018	Chennupalli	N/A	B60K 6/547
2019/0337391	12/2018	Crombez	N/A	B60W 30/18127
2020/0189398	12/2019	Suzuki	N/A	B60T 8/1766
2021/0114464	12/2020	Carbone	N/A	B60L 7/26
2021/0354671	12/2020	Cho	N/A	B60T 8/26
2023/0026361	12/2022	Subramanian	N/A	B60T 8/171

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105292092	12/2016	CN	N/A

Primary Examiner: Zaleskas; John M

Background/Summary

TECHNICAL FIELD

(1) The embodiments described herein are generally directed to braking, and, more particularly, to dynamically splitting braking torque between axles for optimal thermal management.

BACKGROUND

(2) Typically, mobile equipment with axles has a mechanically fixed or constant split ratio for the braking torque that is applied to each axle. Some mobile equipment has advanced features, such as traction control, load-state-based changes, and variable cooling flow split. However, these advanced features are not determined dynamically based on the severity of the braking scenario. In addition, in mobile equipment that employs electric braking, it would be advantageous to reduce the otherwise redundant, oil-cooled brake system, in order to decrease cost, weight, and complexity. This is especially true if the mobile equipment has multiple brake systems, and the brake system of interest is not used as a long-duration, high-energy system, such that transient performance properties are of interest.

(3) U.S. Pat. No. 11,597,283, Chinese Patent No. CN105292092B, and U.S. Patent Pub. No. 2019/0337391 describe systems related to determining how to split braking torque between axles. The present disclosure is directed toward overcoming one or more of the problems discovered by the inventors.

SUMMARY

(4) In an embodiment, a method comprises, by a controller in a mobile equipment that comprises at least two axles that each have at least one braking system, in each of one or more iterations: determining braking torque based on a requested braking from a brake interface; determining braking power based on the determined braking torque and a speed of the mobile equipment; determining a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles; allocating the braking torque between the at least two axles based on the split; and controlling the at least one braking system for each of the at least two axles based on the allocated braking torque.

(5) In an embodiment, a controller, for a mobile equipment that comprises at least two axles that each have at least one braking system, comprises: at least one hardware processor; and software that is configured to, when executed by the at least one hardware processor, in each of one or more iterations, determine braking torque based on a requested braking from a brake interface, determine braking power based on the determined braking torque and a speed of the mobile equipment, determine a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles, allocate the braking torque between the at least two axles based on the split, and control the at least one braking system for each of the at least two axles based on the allocated braking torque.

(6) In an embodiment, a method comprises, by a controller in a mobile equipment that comprises at least two axles that each have at least one braking system, in each of a plurality of iterations: determining braking torque based on a requested braking from a brake interface; determining braking power based on the determined braking torque and a speed of the mobile equipment; determining a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles; adjusting the split based on braking control performed by another system; stabilizing the split when a rate of change in the split with respect to one or more prior ones of the plurality of iterations would violate a constraint; allocating the braking torque between the at least two axles based on the split; and controlling the at least one braking system for each of the at least two axles based on the allocated braking torque.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The details of embodiments of the present disclosure, both as to their structure and operation, may be gleaned in part by study of the accompanying drawings, in which like reference numerals refer to like parts, and in which:

- (2) FIG. 1 illustrates a schematic diagram of mobile equipment, according to an embodiment;
- (3) FIG. 2 illustrates a system for dynamically allocating brake torque between front and rear axles, according to an embodiment;
- (4) FIG. 3 illustrates a process for dynamically allocating brake torque between front and rear axles, according to an embodiment;
- (5) FIG. 4 illustrates an example of a split map, according to an embodiment; and
- (6) FIG. 5 illustrates a block diagram of an example controller, according to an embodiment.

DETAILED DESCRIPTION

(7) The detailed description set forth below, in connection with the accompanying drawings, is intended as a description of various embodiments, and is not intended to represent the only embodiments in which the disclosure may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the embodiments. However, it will be apparent to those skilled in the art that embodiments of the invention can be practiced without these specific details. In some instances, well-known structures and components are shown in simplified form for brevity of description. In addition, for clarity and ease of explanation, some surfaces and details may be omitted in the present description and figures. It should also be understood that the various components illustrated herein are not necessarily drawn to scale. In other words, the features disclosed in various embodiments may be implemented using different relative dimensions within and between components than those illustrated in the drawings.

(8) FIG. 1 illustrates a schematic diagram of mobile equipment **100**, according to an embodiment. Mobile equipment **100** may be any machine with at least two axles and at least one braking system on each of the two axles. Examples of mobile equipment **100** include, without limitation, a wheel loader, dump truck, cold planer, motor grader, road reclaimer, telehandler, tractor-scraper, or the like, as well as ordinary road vehicles (e.g., automobiles, semi-tractors, etc.) and any vehicle with two or more axles that is manufactured by or for Caterpillar Inc. of Irving, Texas, or any of its affiliates or competitors.

(9) As illustrated, mobile equipment **100** comprises a controller **110**, as well as a front axle **120F** and rear axle **120R**. Controller **110** may be one or more electronic control units (ECUs). Controller **110** may be configured to perform only the functions described herein, or may be configured to perform additional functions to those described herein, and may be configured to interact with other controllers onboard mobile equipment **100** (e.g., other electronic control units).

(10) As used herein, a reference numeral with an appended letter or letters will be used to refer to a specific component, whereas the same reference numeral without any appended letters will be used to refer collectively to a plurality of the component or to refer to a generic or arbitrary instance of the component. Thus, for example, the term “axles **120**” refers collectively to front axle **120F** and rear axle **120R**, and the term “axle **120**” may refer to either front axle **120F** or rear axle **120R**.

(11) Each axle **120** comprises at least two wheels **130** on opposing ends of the axle **120**. For example, front axle **120F** comprises a front left wheel **130FL** and a front right wheel **130FR**. Similarly, rear axle **120R** comprises a rear left wheel **130RL** and a rear right wheel **130RR**.

(12) Each wheel **130** may comprise a wheel sensor **135** that measures one or more parameters of the respective wheel **130**. For example, front left wheel sensor **135FL** measures parameter(s) of front left wheel **130FL**, front right wheel sensor **135FR** measures parameter(s) of front right wheel **130FR**, rear left wheel sensor **135RL** measures parameter(s) of rear left wheel **130RL**, and rear right wheel sensor **135RR** measures parameter(s) of rear right wheel **130RR**. Each wheel sensor **135** outputs a signal representing the value of each parameter to controller **110**. The parameter(s) measured by each wheel sensor **135** may comprise wheel speed, ground speed, or the like.

(13) Each wheel **130** may be associated with a respective braking system **140** that applies braking to the respective wheel **130**. For example, front left braking system **140FL** applies braking to front left wheel **130FL**, front right braking system **140FR** applies braking to front right wheel **130FR**, rear left braking system **140RL** applies braking to rear left wheel **130RL**, and rear right braking

system **140RR** applies braking to rear right wheel **130RR**. Braking systems **140** may comprise any suitable type of brakes, including, without limitation, disc brakes, drum brakes, or the like. Each braking system **140** may be controlled by controller **110**. In particular, controller **110** may control the pressure that is applied by front brakes **140FL** and **140FR** and rear brakes **140RL** and **140RR**, according to an allocation of braking torque to front axle **120F** and rear axle **120R**.

(14) Each braking system **140** may comprise a brake sensor **145** that measures one or more parameters of the respective braking system **140**. For example, front left brake sensor **145FL** measures parameter(s) of front left braking system **140FL**, front right brake sensor **145FR** measures parameter(s) of front right braking system **140FR**, rear left brake sensor **145RL** measures parameter(s) of rear left braking system **140RL**, and rear right brake sensor **145RR** measures parameter(s) of rear right braking system **140RR**. Each brake sensor **145** outputs a signal representing the value of each parameter to controller **110**. The parameter(s) measured by each brake sensor **145** may comprise a brake state parameter. The brake state parameter may be a temperature of the respective braking system **140**, such as the temperature of oil in the respective braking system **140**, the temperature at a cooling outlet of the respective braking system **140**, the temperature of a physical component of the respective braking system **140** other than the oil, or the like. More generally, the brake state parameter may be any parameter that indicates or estimates a relative brake state that is representative of the current stress and/or lifecycle health of the respective braking system **140**, and may be a physical measurement, a combination or composite of physical measurements, an estimated score, index, or other measurement output by an algorithm or physics-based model (e.g., that utilizes one or more physical measurements as input), or the like. As an example, higher values of the brake state parameter (e.g., higher temperature values in the event that the brake state parameter includes temperature) may indicate higher stress or strain on brake systems **140** (e.g., due to thermal and/or mechanical loading) than lower values of the brake state parameter. In this case, the value of the brake state parameter may zero below an endurance limit threshold, and increase exponentially above the endurance limit threshold. The brake state parameter may be derived from a plurality of factors related to brake state, such that the total value is greater than the sum of the parts.

(15) In addition to wheel sensors **135** and brake sensors **145**, controller **110** may receive the output of a brake interface **150**. Brake interface **150** may measure the position of a brake pedal, brake level, brake dial, or any other brake mechanism for requesting braking during the operation of mobile equipment **100**. In cases in which mobile equipment **100** is operated locally (i.e., by a local operator, for example, in the cabin of or otherwise onboard mobile equipment **100**), brake interface **150** may sense the requested braking from the brake mechanism (e.g., by a position of the brake mechanism) within the cabin of mobile equipment **100**, as the local operator modulates or otherwise operates it. In cases in which mobile equipment **100** is operated remotely (i.e., by a remote operator offboard mobile equipment **100**), brake interface **150** may by a remote interface that senses the requested braking from the brake mechanism (e.g., by a position of the brake mechanism) within a remote operating terminal, as the remote operator modulates or otherwise operates it, and transmits the requested braking to controller **110** over a wireless communication network. In cases in which mobile equipment **100** is operated autonomously (i.e., with no involvement by a human operator or with supervision by a human operator), the brake mechanism may be omitted or substituted, and the requested braking may be provided directly to controller **110** (e.g., by a separate autonomous driving system) or determined by controller **110** (e.g., if controller **110** implements the autonomous driving system), for example, based on a speed command. In cases in which mobile equipment **100** is operated semi-autonomously (e.g., with some involvement by a human operator), controller **110** may be configured to both receive the requested braking via brake interface **150** during human operation, and directly receive or determine requested braking during autonomous operation. In any case, the requested braking may be represented as an amount of brake torque, an amount of brake power, a target speed over a time interval (e.g., below a current

speed), or other parameter from which an application of braking systems **140** can be derived.

(16) FIG. 2 illustrates a system **200** for dynamically allocating brake torque between front and rear axles **120**, according to an embodiment. System **200** comprises a braking torque determination module **210**, braking power determination module **220**, split map **230**, adjustment module **240**, stabilization module **250**, braking torque allocation module **260**, front braking pressure determination module **270F**, and rear braking pressure determination module **270R**. System **200** may be implemented as one software module or a plurality of software modules that interact with each other (e.g., via inter-process or intra-process communications), and may be executed by a hardware processor or controller **110**. Alternatively, system **200** could be implemented as one or more hardware modules, such as a specific-purpose processor, integrated circuit (IC), application-specific integrated circuit (ASIC), digital signal processor (DSP), field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, and/or the like. As another alternative, system **200** could be implemented as a combination of software and hardware modules. While system **200** is illustrated with a certain arrangement and ordering of modules, system **200** may be implemented with fewer, more, or different modules and a different arrangement and/or ordering of modules.

(17) Braking torque determination module **210** is configured to receive the requested braking from brake interface **150** or an autonomous or semi-autonomous driving system. Braking torque determination module **210** is configured to convert the requested braking into the total value of braking torque to be applied to braking systems **140**, and output this total value of braking torque. In the event that brake interface **150** outputs the requested braking as a position of a torque brake mechanism (e.g., a torque pedal), the total value of braking torque may be calculated as a product or other function of the position and the maximum braking torque value. In the event that brake interface **150** outputs the requested braking as a position of a power brake mechanism (e.g., a power pedal), the total value of braking torque may be calculated as the product of the position and the maximum braking power value, divided by wheel speed, or according to some other function.

(18) Braking power determination module **220** is configured to receive the wheel or ground speed from wheel sensors **135** and the braking torque that is output from braking torque determination module **210**. Braking power determination module **220** converts the wheel or ground speed and the braking torque into a value of braking power, and outputs this value of braking power. The braking power is the potential and kinetic energy that must be removed by mobile equipment **100** per unit time to achieve the determined braking torque.

(19) The braking power, output by braking power determination module **220**, and the brake state parameters (e.g., brake temperatures, or other proxy of brake health or stress), output by brake sensors **145**, may be used as indices into split map **230**, in order to determine the braking torque split. The braking power may be expressed in kilowatts (KW). The brake state parameters may be expressed as a differential between the brake state parameter of the front braking systems (i.e., **140FL** and **140FR**) and the brake state parameter of the rear braking systems (i.e., **140RL** and **140RR**). For example, in an embodiment in which the brake state parameter comprises a brake temperature, the brake temperatures may be expressed as a temperature differential between the temperature of the front braking systems (i.e., **140FL** and **140FR**) and the temperature of the rear braking systems (i.e., **140RL** and **140RR**). The brake state parameter (e.g., temperature) of the front braking systems may be the average of the front braking systems **140FL** and **140FR**, as acquired by brake sensors **145FL** and **145FR**, respectively. Similarly, the brake state parameter (e.g., temperature) of the rear braking systems may be the average of the rear braking systems **140RL** and **140RR**, as acquired by brake sensors **145RL** and **145RR**. The brake state differential (e.g., temperature differential, expressed in degrees of Celsius or Fahrenheit) may be calculated by subtracting the brake state parameter (e.g., temperature) of the rear braking systems from the brake state parameter (e.g., temperature) of the front braking systems.

(20) In an embodiment, the brake state differential may be intentionally biased, such that the

amount of wear is not distributed evenly between axles **120**. This may be useful in cases in which one axle **120** has less wear than the other axle **120** (e.g., prior to implementing the disclosed embodiments), or in which one axle **120** is easier to replace (e.g., more accessible, less costly, etc.) than the other axle **120**. In this case, more wear may be distributed to the axle **120** with less wear (e.g., at least until the wear is more evenly distributed) or that is easier to replace, than to the axle **120** with more wear or that is more difficult to replace.

(21) Split map **230** may comprise or represent a two-dimensional lookup table that determines the braking torque split associated with the input value of braking power and the input value of the brake state parameter (e.g., temperature differential). If an input value of braking power and/or an input value of the brake state parameter are not specifically represented in the lookup table, the braking torque split may be interpolated (e.g., linearly interpolated) based on values of braking power and/or the brake state parameter that are in the lookup table. Alternatively, instead of a lookup table, split map **230** may comprise an algorithm that calculates the braking torque split using the values of braking power and brake state parameter as inputs.

(22) Adjustment module **240** is configured to receive the braking torque split, output by split map **230**, and apply any overrides or other adjustments to the braking torque split output. These adjustments may be responsive to another system, which may perform advanced braking control and which may be onboard mobile equipment **100** or offboard mobile equipment **100**. For example, the other system may implement automatic speed control (e.g., to maintain a set speed), perform automatic braking when an object is detected (e.g., to avoid a collision), respond to offboard requests (e.g., to maintain speed, decelerate, accelerate, etc., to anticipate a required power, to maintain a particular parameter for a particular duration, etc.), and/or the like. In this case, the determination of this other system, regarding the braking torque split or other parameter that affects the braking torque split, may override or otherwise cause an adjustment to the braking torque split determined by split map **230**. Since the severity and duration of braking is known in advance, the actions by the other system can be more precise, whereas future input by an operator is unknown. For example, the brake torque split may be adjusted based on braking control performed by this other system. Such an embodiment is appropriate when unhindered operation of the other system is more important or appropriate, in the moment, than the health of braking systems **140**. For instance, when one axle **120** loses traction and begins sliding, the other system may allocate more braking torque to the other axle **120** in order to maintain control. This traction control should override split map **230**, since maintaining traction is more important than the health of brake systems **140** in the moment. In an alternative embodiment, adjustment module **240** may be omitted. It should be understood that, even in embodiments which include adjustment module **240**, adjustment module **240** may not always adjust the braking torque split. Rather, adjustment module **240** will only adjust the braking torque split when dictated by the other system.

(23) Stabilization module **250** is configured to receive the braking torque split, output by split map **230** and potentially adjusted by adjustment module **240**, and stabilize the braking torque split by limiting the rate of change in the value of the braking torque split, with respect to previously determined brake torque splits, such that the rate of change does not violate a constraint (e.g., by constraining the rate of change to below a threshold value). In other words, stabilization module **250** prevents the braking torque split from bouncing between different values over a short period of time. In an embodiment, stabilization module **250** implements proportional-integral-derivative (PID) control, which employs feedback to gradually correct the error between the requested value of the braking torque split and the actual value of the braking torque split. In an alternative embodiment, stabilization module **250** may be omitted.

(24) Braking torque allocation module **260** is configured to receive the braking torque, output by braking torque determination module **210**, and the braking torque split, output by stabilization module **250**. Braking torque allocation module **260** allocates the braking torque between the front and rear braking systems **140** according to the braking torque split. For example, if the braking

torque split is 40%, then 40% of the braking torque, output by braking torque determination module **210**, is allocated to the front braking systems **140FL** and **140FR**, and 60% of the braking torque, output by braking torque determination module **210**, is allocated to the rear braking systems **140RL** and **140RR**.

(25) Braking torque allocation module **260** outputs the front braking torque to front braking pressure determination module **270F**, and outputs the rear braking torque to rear braking pressure determination module **270R**. Each braking pressure determination module **270** converts the received braking torque into a braking pressure, and outputs the braking pressure to the respective braking systems **140**. For example, front braking pressure determination module **270F** outputs the front braking pressure to front braking systems **140FL** and **140FR**, and rear braking pressure determination module **270R** outputs the rear braking pressure to rear braking systems **140RL** and **140RR**. Each respective braking system **140** then applies the received braking pressure to the respective wheel **130**.

(26) FIG. 3 illustrates a process **300** for dynamically allocating brake torque between front and rear axles **120**, according to an embodiment. It should be understood that process **300** may be implemented using system **200**. Process **300** may be executed over a plurality of iterations, for example, starting from when system **200** is activated (e.g., when mobile equipment **100** is turned on) and ending when system **200** is deactivated (e.g., when mobile equipment **100** is turned off). Process **300** may be executed in real time as mobile equipment **100** is operating. As used herein, the terms “real time” or “real-time” encompass both a response that occurs simultaneously with an input event, as well as a response that occurs some time after the input event due to ordinary latencies inherent in processing, communications, and/or the like.

(27) While process **300** is illustrated with a certain arrangement and ordering of subprocesses, process **300** may be implemented with fewer, more, or different subprocesses and a different arrangement and/or ordering of subprocesses. In addition, it should be understood that any subprocess, which does not depend on the completion of another subprocess, may be executed before, after, or in parallel with that other independent subprocess, even if the subprocesses are described or illustrated in a particular order.

(28) In subprocess **305**, process **300** determines whether or not to continue. For example, process **300** may continue for as long as system **200** is active. System **200** may be configured to be enabled and disabled at the discretion of a human operator (e.g., via a button or other input in the cabin of mobile equipment **100** and/or at a remote terminal) and/or autonomous driving system (e.g., via a control signal), and/or may be automatically enabled when mobile equipment **100** is turned on and remain active until mobile equipment **100** is turned off or system **200** is disabled. Thus, system **200** can be activated or deactivated as needed or desired. When determining that process **300** is to continue (i.e., “Yes” in subprocess **305**), process **300** proceeds to subprocess **310**. Otherwise, when determining that process **300** is not to continue (i.e., “No” in subprocess **305**), process **300** ends.

(29) In subprocess **310**, the braking torque is determined based on the requested braking from brake interface **150**, as acquired in real time. Subprocess **310** may be implemented by braking torque determination module **210**. Therefore, any description of braking torque determination module **210** applies equally to subprocess **310**, and vice versa.

(30) In subprocess **320**, the braking power is determined based on the braking torque, determined in subprocess **310**, and the wheel or ground speed (e.g., acquired from wheel sensors **135** in real time) of mobile equipment **100**. Subprocess **320** may be implemented by braking power determination module **220**. Therefore, any description of braking power determination module **220** applies equally to subprocess **320**, and vice versa.

(31) In subprocess **330**, the optimal split of braking torque between front and rear braking systems **140** is determined based on the braking power, determined in subprocess **320**, and the brake state parameter (e.g., determined from the output of brake sensors **145**, acquired in real time). Subprocess **330** may be implemented by split map **230**. Therefore, any description of split map **230**

applies equally to subprocess **330**, and vice versa.

(32) In subprocess **340**, the braking torque split, determined in subprocess **330**, may be adjusted based on the operation of another braking control system. Subprocess **340** may be implemented by adjustment module **240**. Therefore, any description of adjustment module **240** applies equally to subprocess **340**, and vice versa. In an alternative embodiment, subprocess **340** may be omitted.

(33) In subprocess **350**, the braking torque split, determined in subprocess **330** and potentially adjusted in subprocess **340**, may be limited for stabilization. Subprocess **350** may be implemented by stabilization module **250**. Therefore, any description of stabilization module **250** applies equally to subprocess **350**, and vice versa. In an alternative embodiment, subprocess **350** may be omitted.

(34) In subprocess **360**, the braking torque, determined in subprocess **310**, is allocated between front axle **120F** and rear axle **120R** based on the braking torque split, output by subprocess **350** and/or input to subprocess **360**. Subprocess **360** may be implemented by braking torque allocation module **260**. Therefore, any description of braking torque allocation module **260** applies equally to subprocess **360**, and vice versa.

(35) In subprocess **370**, the front braking pressure for front braking systems **140FL** and **140FR** and the rear braking pressure for rear braking systems **140RL** and **140RR** are determined based on the allocations, determined in subprocess **360**. Subprocess **370** may be implemented by braking pressure determination modules **270**. Therefore, any description of front braking pressure determination module **270F** and/or rear braking determination module **270R** applies equally to subprocess **370**, and vice versa.

(36) In subprocess **380**, braking systems **140** are controlled in accordance with the braking pressures determined in subprocess **370**. For example, controller **110** may apply the front braking pressure, determined in subprocess **370**, to front braking systems **140FL** and **140FR**. Similarly, controller **110** may apply the rear braking pressure, determined in subprocess **370**, to rear braking systems **140RL** and **140RR**.

(37) FIG. 4 illustrates an example of a split map **230**, according to an embodiment. In this example, each row represents a value of the temperature differential between front and rear braking systems **140**, and each column represents a value of braking power. While the brake temperature differential is used to represent the brake state parameter in this example, the brake state parameter, including a brake state parameter that comprises a temperature, could be represented in an alternative manner. For example, actual brake temperatures often lag and may not be completely indicative of brake stress. Thus, in an alternative embodiment, a proxy, such as a thermal estimator (e.g., generated by an algorithm or physics-based model based on one or more physical measurements, such as current brake temperatures), could be used as the brake state parameter for split map **230**. More generally, any parameter indicative of the state of braking systems **140** may be used as an input, in place of the brake temperature differential, to split map **230**.

(38) In the illustrated example, the value in each cell in split map **230** represents the percentage of braking torque that should be allocated to front braking systems **140FL** and **140FR** for a given pairing of braking power and brake state parameter. It should be understood that the remaining percentage of braking torque should be allocated to rear braking systems **140RL** and **140RR**. Thus, for example, at a brake temperature differential value of -5°C . (e.g., determined from brake sensors **145**) and a braking power of 500 kW (e.g., determined by braking power determination module **220** in subprocess **320**), 40% of the braking torque (e.g., determined by braking torque determination module **210** in subprocess **310**) may be allocated to front braking systems **140FL** and **140FR**, and 60% of the braking torque may be allocated to rear braking systems **140RL** and **140RR**.

(39) Allocation values for pairings of braking power and the brake state parameter that are not explicitly represented in split map **230** may be interpolated. For example, at a brake temperature differential value of 10°C . and a braking power of 250 KW, the value of the braking torque split may be linearly interpolated as 12.5%.

(40) As mentioned above, split map **230** could alternatively be implemented as an algorithm, instead of a lookup table. In either case, split map **230** may comprise any number of dimensions, including one dimension or more than the two illustrated dimensions. In addition, split map **230** could comprise a different set of dimensions than those illustrated. For example, other parameters may be used instead of, or in addition to, braking power and/or brake temperature differential. In particular, as mentioned above, the brake temperature differential may be replaced with one or more other parameters/dimensions indicative of the relative state of braking systems **140**. As mentioned previously, split map **230** may be intentionally biased to distribute more braking torque—and therefore, more wear—to one axle **120** than the other axle **120**, for example, as a means to balance the state of health of previously unbalanced brake systems **140** over time, or to create and/or maintain an imbalance in the state of health that increases the relative wear to an axle **120** that is easier to replace.

(41) Split map **230** may be determined initially using a simulation with a digital twin of mobile equipment **100**. In particular, the braking torque split may be optimized across all possible or feasible values in each dimension (e.g., braking power and brake state parameter). These optimal braking torque splits may then be represented in split map **230**, which may be loaded into memory of controller **110**. The optimal braking torque split for each combination of dimension values represents the braking torque split that evenly distributes damage across all braking systems **140** in the scenario represented by that combination of dimension values, such that braking systems **140** wear evenly over time. In an embodiment, split map **230** may contain non-linear transitions between braking torque splits.

(42) Notably, the initial split map **230** may not be ideally balanced for a particular mobile equipment **100**, especially if brake temperature is used as the brake state parameter, since the brake temperature is a lagging indicator. However, split map **230** may still provide a suitable braking torque split at the start of a braking event. Even in the event that the brake state parameter is a lagging indicator such as temperature, the values of the brake state parameter from brake sensors **145** will eventually catch up, such that the braking torque split can be fine-tuned over the remainder of the braking event.

(43) In an embodiment, the initial split map **230**, determined via simulation, may be adjusted over time using feedback from operation of the actual mobile equipment **100**. For example, controller **110** may continually (e.g., in real time or periodically) assess real-time feedback from onboard sensors (e.g., brake sensors **145**) to determine how well the current split map **230** is performing in terms of the objective (e.g., evenly distributing brake wear, temperature, or other health imbalances across axles **120**). If the current split map **230** is underperforming, controller **110** may adjust split map **230** (e.g., adjust the braking torque split values in split map **230** for one or more combinations of dimensions) to improve performance. Controller **110** may modify split map **230** directly, or may adjust an offset value that is applied to the values of the braking torque split that are output by split map **230**. Thus, split map **230** can evolve over time to account for the actual operating conditions of mobile equipment **100**. In this case, the split map **230** for two different mobile equipment **100** of the same model may start out identical but begin to differ over time.

(44) FIG. 5 illustrates a block diagram of an example controller **110**, according to an embodiment. Controller **110** may implement process **300**, for example, by executing the modules of system **200**. In this embodiment of controller **110**, system **200** is implemented as computer-executable software modules.

(45) Controller **110** may comprise one or more processors **510**. Processor(s) **510** may comprise a central processing unit (CPU). Additional processors may be provided, such as a graphics processing unit (GPU), an auxiliary processor to manage input/output, an auxiliary processor to perform floating-point mathematical operations, a special-purpose microprocessor having an architecture suitable for fast execution of signal-processing algorithms (e.g., digital-signal processor), a subordinate processor (e.g., back-end processor), an additional microprocessor or

controller for dual or multiple processor systems, and/or a coprocessor. Such auxiliary processors may be discrete processors or may be integrated with a main processor **510**. Examples of processors which may be used with controller **110** include, without limitation, any of the processors (e.g., Pentium™, Core i7™, Xeon™, etc.) available from Intel Corporation of Santa Clara, California, any of the processors available from Advanced Micro Devices, Incorporated (AMD) of Santa Clara, California, any of the processors (e.g., A series, M series, etc.) available from Apple Inc. of Cupertino, any of the processors (e.g., Exynos™) available from Samsung Electronics Co., Ltd., of Seoul, South Korea, any of the processors available from NXP Semiconductors N.V. of Eindhoven, Netherlands, and/or the like.

(46) Processor **510** may be connected to a communication bus **505**. Communication bus **505** may include a data channel for facilitating information transfer between storage and other peripheral components of controller **110**. Furthermore, communication bus **505** may provide a set of signals used for communication with processor **510**, including a data bus, address bus, and/or control bus (not shown). Communication bus **505** may comprise any standard or non-standard bus architecture such as, for example, bus architectures compliant with industry standard architecture (ISA), extended industry standard architecture (EISA), Micro Channel Architecture (MCA), peripheral component interconnect (PCI) local bus, standards promulgated by the Institute of Electrical and Electronics Engineers (IEEE) including IEEE 488 general-purpose interface bus (GPIB), IEEE 696/S-100, and/or the like.

(47) Controller **110** may comprise main memory **515**. Main memory **515** provides storage of instructions and/or other data for software executing on processor **510**, such as modules **210-270** of system **200**. It should be understood that instructions stored in the memory and executed by processor **510** may be written and/or compiled according to any suitable language, including without limitation C/C++, Java, JavaScript, Perl, Python, Visual Basic, .NET, and the like. Main memory **515** is typically semiconductor-based memory such as dynamic random access memory (DRAM) and/or static random access memory (SRAM). Other semiconductor-based memory types include, for example, synchronous dynamic random access memory (SDRAM), Rambus dynamic random access memory (RDRAM), ferroelectric random access memory (FRAM), and the like, including read only memory (ROM).

(48) Controller **110** may comprise secondary memory **520**. Secondary memory **520** is a non-transitory computer-readable medium having instructions and/or other data for software, such as modules **210-270** of controller **110**, stored thereon. In this description, the term “computer-readable medium” is used to refer to any non-transitory computer-readable storage media used to provide computer-executable code and/or other data to or within controller **110**. The computer software stored on secondary memory **520** is read into main memory **515** for execution by processor **510**. Secondary memory **520** may include, for example, semiconductor-based memory, such as programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable read-only memory (EEPROM), and flash memory (block-oriented memory similar to EEPROM).

(49) Controller **110** may comprise an input/output (I/O) interface **535**. I/O interface **535** provides an interface between one or more components of controller **110** and one or more input and/or output devices. For example, I/O interface **535** may receive the output of one or more sensors (e.g., wheel sensors **135**, brake sensors **145**, pedal sensor **150**, etc.), and/or output control signals to one or more components of mobile equipment **100** (e.g., braking systems **140**).

(50) Controller **110** may comprise a communication interface **540**. Communication interface **540** allows software to be transferred between controller **110** and external devices, networks, or other information sources. For example, instructions and/or other data may be transferred to controller **110**, over one or more networks, from a network server via communication interface **540**. Examples of communication interface **540** include a built-in network adapter, network interface card (NIC), Personal Computer Memory Card International Association (PCMCIA) network card,

card bus network adapter, wireless network adapter, Universal Serial Bus (USB) network adapter, modem, a wireless data card, a communications port, an infrared interface, an IEEE 1394 fire-wire, and any other device capable of interfacing controller **110** with a network or another computing device. Communication interface **540** preferably implements industry-promulgated protocol standards, such as Ethernet IEEE 802 standards, Fiber Channel, digital subscriber line (DSL), asynchronous digital subscriber line (ADSL), frame relay, asynchronous transfer mode (ATM), integrated digital services network (ISDN), personal communications services (PCS), transmission control protocol/Internet protocol (TCP/IP), serial line Internet protocol/point to point protocol (SLIP/PPP), and so on, but may also implement customized or non-standard interface protocols as well.

(51) Software transferred via communication interface **540** is generally in the form of electrical communication signals **555**. These signals **555** may be provided to communication interface **540** via a communication channel **550** between communication interface **540** and an external system **545**. In an embodiment, communication channel **550** may be a wired or wireless network, or any variety of other communication links. Communication channel **550** carries signals **555** and can be implemented using a variety of wired or wireless communication means including wire or cable, fiber optics, conventional phone line, cellular phone link, wireless data communication link, radio frequency (“RF”) link, or infrared link, just to name a few.

(52) Computer-executable code is stored in main memory **515** and/or secondary memory **520**. Computer-executable code can also be received from an external system **545** via communication interface **540** and stored in main memory **515** and/or secondary memory **520**. Such computer-executable code, when executed by processor(s) **510**, enable controller **110** to perform the various functions of the disclosed embodiments.

INDUSTRIAL APPLICABILITY

(53) Typically, mobile equipment with axles has a mechanically fixed or constant split ratio for the braking torque that is applied to each axle. Disclosed embodiments dynamically adjust the braking torque ratio between axles **120** based on real-time parameter values, such as requested braking power and a real-time brake state parameter (e.g., brake temperatures), to more evenly distribute wear or other health imbalances across brake systems **140** of mobile equipment **100**. Accordingly, disclosed embodiments may extend the longevity of brake systems **140**, reduce the costs and frequency of maintenance of mobile equipment **100**, and/or the like.

(54) Furthermore, disclosed embodiments enable a reduction in redundant oil-cooled brake systems **140**, especially in scenarios in which brake systems **140** are not used as long-duration high-energy systems. In particular, there are two bounding load case scenario extremes. At the first extreme, a braking event occurs with high power and short duration (i.e., low energy). The inventors have recognized that, in this case, the thermal mass of the system is the dominant factor. Therefore, the braking torque should be split in proportion to the combination of the energy dissipation capabilities of the thermal mass and the oil cooling flow to brake systems **140**. At the second extreme, a downhill braking event occurs at fixed speed with low power and long duration (i.e., high energy). The inventors have recognized that, in this case, the thermal mass of the system is not a relevant factor. Therefore, the braking torque should be split in proportion to just the oil cooling flow to brake systems **140**. Accordingly, split map **230** splits the braking torque differently along the spectrum between these two extremes to achieve the optimal split based on the real-time physical properties of mobile equipment **100**.

(55) In addition, between these two extremes, the transition between splits in split map **230** may be non-linear. Thus, offboard simulations may be run to develop an initial split map **230** that is based on actual conditions and which has non-linear transitions between splits.

(56) It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. Aspects described in connection with one embodiment are intended to be able to be used with the other embodiments. Any explanation in

connection with one embodiment applies to similar features of the other embodiments, and elements of multiple embodiments can be combined to form other embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages.

(57) The preceding detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. The described embodiments are not limited to usage in conjunction with a particular type of mobile equipment. Hence, although the present embodiments are, for convenience of explanation, depicted and described as being implemented in a two-axle mobile equipment, it will be appreciated that it can be implemented in mobile equipment with a different number of axles and various other types of machines with brakes, and in various other systems and environments. Furthermore, there is no intention to be bound by any theory presented in any preceding section. It is also understood that the illustrations may include exaggerated dimensions and graphical representation to better illustrate the referenced items shown, and are not considered limiting unless expressly stated as such.

Claims

1. A method comprising, by a controller in a mobile equipment that comprises at least two axles that each have at least one braking system, in each of one or more iterations: determining braking torque based on a requested braking from a brake interface; determining braking power based on the determined braking torque and a speed of the mobile equipment; determining a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles; allocating the braking torque between the at least two axles based on the split; and controlling the at least one braking system for each of the at least two axles based on the allocated braking torque.
2. The method of claim 1, further comprising, by the controller, in each of the one or more iterations, adjusting the split based on braking control performed by another system.
3. The method of claim 1, wherein the one or more iterations are a plurality of iterations, and wherein the method further comprises, by the controller, in at least one of the plurality of iterations, constraining a rate of change in a value of the split with respect to one or more prior ones of the plurality of iterations.
4. The method of claim 1, wherein controlling the at least one braking system on each of the at least two axles based on the allocated braking torque comprises: determining a first brake pressure based on the allocated braking torque for a first one of the at least two axles; determining a second brake pressure based on the allocated braking torque for a second one of the at least two axles; controlling the at least one braking system for the first axle based on the first brake pressure; and controlling the at least one braking system for the second axle based on the second brake pressure.
5. The method of claim 1, wherein the at least two axles comprise a front axle and a rear axle.
6. The method of claim 5, wherein each of the front axle and the rear axle comprises two wheels, wherein the mobile equipment comprises a braking system on each of the two wheels on each of the front and rear axles, and wherein controlling the at least one braking system for each of the at least two axles comprises controlling the braking system on each of the two wheels on each of the front and rear axles.
7. The method of claim 1, wherein determining the split of the braking torque between the at least two axles comprises looking up a value of the split in a split map using the braking power and the brake state parameters as inputs.
8. The method of claim 7, wherein the brake state parameters are represented in the inputs as a single differential between a first brake state parameter of the at least one braking system on a first one of the at least two axles and a second brake state parameter of the at least one braking system on a second one of the at least two axles.

9. The method of claim 8, wherein each of the first brake state parameter and the second brake state parameter is a brake temperature.
 10. The method of claim 7, further comprising, by the controller, updating the split map based on real-time feedback from one or more onboard sensors of the mobile equipment.
 11. The method of claim 1, wherein determining the split of the braking torque between the at least two axles comprises calculating a value of the split using an algorithm with the braking power and brake state parameters as inputs to the algorithm.
 12. The method of claim 11, wherein the brake state parameters are represented in the inputs as a single differential between a first brake state parameter of the at least one braking system on a first one of the at least two axles and a second brake state parameter of the at least one braking system on a second one of the at least two axles.
 13. The method of claim 12, wherein each of the first brake state parameter and the second brake state parameter is a brake temperature.
 14. The method of claim 11, further comprising, by the controller, updating the algorithm based on real-time feedback from one or more onboard sensors of the mobile equipment.
 15. A controller for a mobile equipment that comprises at least two axles that each have at least one braking system, the controller comprising: at least one hardware processor; and software that is configured to, when executed by the at least one hardware processor, in each of one or more iterations, determine braking torque based on a requested braking from a brake interface, determine braking power based on the determined braking torque and a speed of the mobile equipment, determine a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles, allocate the braking torque between the at least two axles based on the split, and control the at least one braking system for each of the at least two axles based on the allocated braking torque.
 16. The controller of claim 15, wherein the software is further configured to, in each of the one or more iterations, adjust the split based on braking control performed by another system.
 17. The controller of claim 15, wherein the one or more iterations are a plurality of iterations, and wherein the software is further configured to, in at least one of the plurality of iterations, constrain a rate of change in a value of the split with respect to one or more prior ones of the plurality of iterations.
 18. The controller of claim 15, further comprising a non-transitory computer-readable medium storing a split map thereon, wherein determining the split of the braking torque between the at least two axles comprises looking up a value of the split in the split map using the braking power and brake state parameters as inputs.
 19. A mobile equipment comprising: at least two axles, wherein each of the at least two axles has at least one braking system; and the controller of claim 15.
 20. A method comprising, by a controller in a mobile equipment that comprises at least two axles that each have at least one braking system, in each of a plurality of iterations: determining braking torque based on a requested braking from a brake interface; determining braking power based on the determined braking torque and a speed of the mobile equipment; determining a split of the braking torque between the at least two axles based on the braking power and brake state parameters of the at least one braking system on each of the at least two axles; adjusting the split based on braking control performed by another system; stabilizing the split when a rate of change in the split with respect to one or more prior ones of the plurality of iterations would violate a constraint; allocating the braking torque between the at least two axles based on the split; and controlling the at least one braking system for each of the at least two axles based on the allocated braking torque.
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